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Enhancing DDoS Resilience in Government Voter Systems: Extending Google Project Shield with Real-Time Anomaly-Aware L4–L7 Adaptive Filtering

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Abstract

As digital election systems become more interconnected and internet-facing, ensuring their availability and resilience against cyberattacks, especially Distributed Denial of Service (DDoS) attacks has become critically important. These systems are particularly vulnerable during peak periods such as voter registration deadlines and election days, where even brief disruptions can damage public trust and interfere with democratic processes. This project investigates the integration of real-time anomaly-aware Layer 4 (L4) and Layer 7 (L7) filtering into Google Project Shield as a means of strengthening the protection of government voter systems. The proposed enhancement uses adaptive baseline modeling to continuously learn normal traffic behavior and respond intelligently to abnormal patterns, removing the need for manual rule-setting or physical mitigation hardware. Designed to operate in a lightweight, cloud-native architecture, this solution is intended to be easily deployable across various infrastructures, making it especially suitable for election commissions in resource-limited settings. The research also explores the system's technological context, recent innovations in multi-layer filtering, and the vulnerabilities faced by public election platforms, while outlining the methodology used to assess the solution's practicality and impact.

By addressing these challenges, the proposed system aims to deliver a scalable, automated, and cost-effective approach to modern DDoS mitigation in the public sector.

Index Terms

DDoS Protection, Election Security, Layer 4–7 Filtering, Anomaly Detection, Google Project Shield, Real-Time Traffic Analysis.

1. Introduction

Government election systems are increasingly vulnerable to cyberattacks, particularly Distributed Denial of Service (DDoS) attacks that are timed to disrupt services during critical periods such as voter registration and election days. Tools like Google Project Shield provide strong Layer 7 (application-level) protection but lack native support for Layer 4 (transport-level) defense, leaving APIs and voter-facing platforms exposed to volumetric threats. This research identifies the risks posed by this gap, especially for resource-constrained election agencies, and proposes enhancing the resilience of existing systems through real-time anomaly-aware adaptive filtering. This approach enables traffic baselining during normal periods and automatic rule deployment during anomalies, ensuring availability and reliability without additional physical infrastructure.

The study is structured into multiple analytical sections to explore technologies and improvements related to this approach. The first domain discusses the core feature real-time anomaly-aware adaptive filtering and highlights its role in accurately identifying and mitigating DDoS traffic. The second domain reviews current cloud-based DDoS mitigation technologies, including their reliance on L7 filtering and their limitations in addressing transport-layer threats. The third domain explores improvements found in systems that use similar adaptive filtering techniques, demonstrating how their architecture and deployment strategies can inform more effective DDoS protection in government election infrastructure. These are followed by a comparative analysis of similar systems, a literature review of recent research in multi-layer filtering, and a methodology section outlining the intended users, sampling techniques, and data collection methods for evaluating the approach's viability.

innovations relevant to defending government voter systems against Distributed Denial of Service (DDoS) attacks. It focuses on real-time anomaly-aware filtering approaches, the limitations of existing cloud-based mitigation tools, and techniques from similar systems in adjacent sectors. With specific emphasis on Layer 4 and Layer 7 protection, this review identifies how lightweight, cloud-native solutions can improve resilience without requiring complex hardware deployment. The review is structured across key themes, including adaptive filtering methods, the role of current mitigation platforms, architecture designs that support flexible deployment, and comparative studies of related technologies in industrial, IoT, and public infrastructure environments.

2.1 Real-Time Anomaly-Aware Adaptive Filtering

2. Literature Review

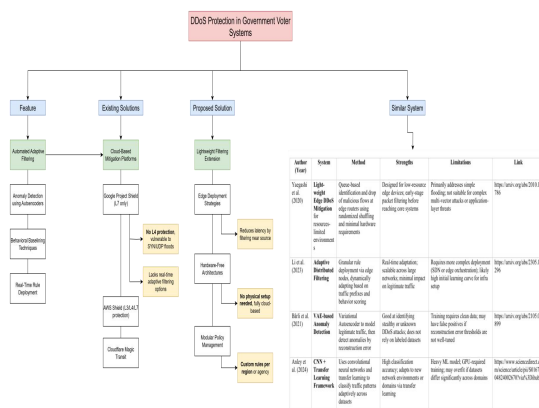


Figure 1: Literature Review Map

This literature review explores the technological foundations, gaps, and

The core innovation of the proposed system lies in its use of real-time anomaly-aware adaptive filtering to defend against DDoS attacks targeting government election systems. Unlike static filters or rule-based detection techniques, anomaly-aware filtering systems are capable of learning baseline traffic behavior and identifying subtle deviations that may indicate the onset of an attack. This dynamic approach enables the system to respond to new or evolving threats in real-time, which is crucial for maintaining availability during critical election periods such as voter registration and online polling. As government systems face increasingly complex threats across multiple OSI layers, the

integration of adaptive filtering offers a more intelligent and resilient solution.

Recent research supports the efficacy of this approach. (Li et al., 2023) introduced an adaptive distributed filtering framework designed to handle DDoS traffic at an Internet scale. The system dynamically adapts filtering granularity based on changing traffic behavior and network load. It can make rule adjustments in real-time, applying them across edge nodes with minimal delay. This distributed architecture ensures the fast and localized mitigation of malicious traffic without affecting legitimate user access, a key consideration for public voting platforms that must remain transparent and accessible. Their model demonstrated significant improvements in filtering accuracy, system responsiveness, and scalability qualities necessary for national-level election systems.

In addition to distribution and responsiveness, anomaly detection must also be accurate. (Wei et al., 2023) presented a novel model using an LSTM-Autoencoder architecture to detect anomalies in multivariate time-series data generated by network traffic. Their model was trained to recognize legitimate user behavior and detect abnormalities based on reconstruction error, even when the anomalies were subtle or occurred within legitimate-looking packets. This kind of detection is particularly effective for identifying stealthy or low-rate DDoS attacks that traditional systems might miss. Their experimental results showed detection accuracy above 99%, which underscores the importance of deep

learning-based techniques for high-stakes environments like elections.

Complementing these advances, hybrid detection systems that combine signature-based and behavior-based methods have also proven effective. (Hatef et al., 2020) explored a system that leverages both supervised and unsupervised learning, integrating signature detection (via Snort) with clustering and classification models such as decision trees and Naïve Bayes. The result was a flexible system capable of real-time decision-making without relying on predefined rules alone. This approach is particularly suitable for government environments where new attack vectors can appear without warning, and where human oversight may be limited due to resource constraints.

(Barbhuiya et al., 2018) further illustrated the value of anomaly-aware filtering through the RADS system, designed for cloud data centers. RADS used a sliding-window time-series approach combined with one-class classification to detect DDoS and crypto-mining anomalies in live traffic. Deployed in real-world environments, the system maintained a false-positive rate below 3% while achieving 90–95% detection accuracy. This research reinforces the viability of deploying real-time adaptive systems in production networks, with minimal overhead and strong detection capabilities.

The impact of implementing real-time anomaly-aware adaptive filtering in

government election systems is significant. Such systems reduce false positives by relying on behavior rather than fixed signatures or thresholds, ensuring that legitimate users are not mistakenly blocked during traffic surges. They also provide early warning capabilities by detecting behavioral deviations before volumetric peaks cause damage. Because these systems are software-based and cloud-native, they eliminate the need for expensive physical infrastructure, making them accessible to lower-resource government agencies. Most importantly, they ensure uninterrupted access to democratic services during high-risk periods, enhancing public trust in digital governance and reinforcing electoral integrity.

Overall, the integration of real-time anomaly-aware adaptive filtering into government voter platforms fills a critical gap in existing DDoS defense frameworks. Supported by advances in machine learning, distributed systems, and hybrid detection strategies, this feature offers a scalable and responsive foundation for securing public digital infrastructure. By enabling intelligent, autonomous mitigation at both L4 and L7, the system can neutralize complex, layered DDoS attacks that would otherwise compromise system availability and public trust during elections.

2.2 Cloud-Based DDoS Mitigation and Layer-Limited Filtering

Contemporary DDoS mitigation is primarily dominated by commercial

cloud-based platforms, including AWS Shield, Akamai Kona Site Defender, and Cloudflare Magic Transit. These services offer layered protection with capabilities such as global scrubbing centers, IP reputation filtering, and rate-limiting. However, they are typically designed for enterprise-level clients and are often costly, putting them beyond the reach of small public-sector entities such as local or regional election commissions. These systems are generally strong at mitigating Layer 7 (L7) threats, focusing on filtering malicious HTTP or API-level traffic using technologies such as Web Application Firewalls (WAFs), CDN caching, and behavioral heuristics (Ali et al., 2023).

One notable solution available to election-focused websites is Google Project Shield, which provides free L7 DDoS protection. Project Shield leverages Google's global infrastructure to absorb and mitigate common application-layer attacks such as HTTP GET/POST floods. While effective in addressing high-frequency bot traffic at the L7 layer, the platform lacks integrated protection for Layer 4 (L4) attacks such as SYN floods, UDP amplification, and TCP exhaustion, which are capable of overwhelming systems before traffic reaches the application level (Jain et al., 2022). This limitation makes election systems particularly vulnerable, as attackers often use a combination of transport and application-layer vectors to bypass incomplete defense architectures.

In lower-resource government environments, the absence of full-stack DDoS protection has led to a reliance on static and manual defense strategies.

These include fixed IP blacklists, manual threshold-based rate limiting, and intrusion detection systems that lack context-aware learning capabilities. Research by Sandhya and Shalinie (2022) found that such strategies, while simple to deploy, struggle to differentiate between legitimate traffic spikes and malicious floods, especially during high-demand periods like voter registration deadlines. As a result, these systems often produce high false-positive rates, inadvertently blocking real users while still failing to prevent sophisticated attacks.

Layer 4 defenses are further complicated by their hardware requirements. Technologies such as deep packet inspection (DPI) and protocol-based rate-limiting are typically implemented in physical appliances like firewalls or specialized scrubbing devices. These systems are not only expensive but also require ongoing maintenance and expert oversight. For election commissions in developing countries, acquiring and integrating these hardware-based solutions poses significant financial and logistical challenges (Singh et al., 2023). Moreover, during high-scale DDoS events, hardware-based defenses can become bottlenecks themselves, struggling to cope with large packet volumes and degrading overall performance.

The evolving nature of DDoS attacks exacerbates the issue. Modern attack campaigns employ multi-vector approaches, often beginning with transport-layer floods to create latency and confusion, followed by application-layer abuse to break critical functions. Akamai's recent State of the Internet

report highlighted a 39% increase in multi-vector attacks between 2022 and 2023, noting that over 50% of observed DDoS events now target both L4 and L7 simultaneously (Kumar & Bhatia, 2023). This hybrid approach renders single-layer protections ineffective, necessitating comprehensive, layered defense strategies that can dynamically respond to different vectors in real-time.

Current cloud solutions, while technically advanced, lack the affordability and flexibility required by public election systems. The inability to filter L4 traffic within free or low-cost platforms like Google Project Shield leaves critical endpoints such as voter registration APIs and HTTPS login portals vulnerable to disruption. In contexts where fair and timely access to online election services is essential, even brief downtime can undermine public trust and electoral integrity. There is thus a growing demand for cloud-native, real-time filtering mechanisms that provide integrated L4 and L7 protection without the need for expensive hardware or high subscription costs.

2.3 Lightweight Real-Time Filtering Extension for Project Shield

To address the existing gaps in Google Project Shield's protection architecture, particularly at Layer 4 (L4), this research proposes a lightweight, cloud-native adaptive filtering extension. The proposed solution operates alongside Project Shield by incorporating real-time behavioral analysis to filter volumetric traffic attacks before they reach application endpoints. While Project

Shield effectively mitigates L7 attacks like HTTP GET floods, it lacks coverage for L4 attacks such as TCP SYN floods or UDP amplification. This gap exposes voter registration APIs and HTTPS-based services to transport-layer threats that can saturate bandwidth and interrupt electoral processes (Sharma et al., 2022).

The solution's architecture is built around the concept of behavioral baselining. By monitoring network traffic during idle periods, the system learns common packet behaviors, including average payload size, source IP frequency, connection initiation patterns, and session persistence. When deviations from these patterns are detected during peak activity, especially those that deviate beyond an adaptive confidence interval, the filtering engine automatically applies mitigation actions. These actions may include throttling the number of concurrent sessions from specific IP addresses, dropping anomalous payloads, or redirecting questionable traffic to a sandboxed verification layer (Sabeel et al., 2023). This approach emphasizes proactive mitigation, as opposed to reactive manual tuning that is often too slow for high-intensity DDoS campaigns.

One of the system's critical innovations is its ability to function entirely in the cloud without hardware dependence. It integrates seamlessly with Google's existing infrastructure and supports deployment to CDN edge nodes, where the filtering rules are applied with minimal latency. The model's modularity allows different election commissions to tune detection thresholds and response actions based on local network behavior. For instance, countries with larger voter

populations may set broader thresholds to avoid false positives, while smaller jurisdictions might prioritize precision in anomaly detection. This flexibility enables tailored protection for governments of various scales and technical capacities (Singh & Dutta, 2022).

The mitigation logic is built using lightweight rules that are compiled from learned traffic behavior and can be deployed instantly across distributed nodes. Because the system updates its rule set in response to new patterns rather than relying on static configurations it stays ahead of attackers who evolve their techniques. Moreover, the filtering model uses unsupervised learning techniques such as clustering or contractive autoencoders, which are ideal for environments where attack types are unpredictable or insufficiently labeled (Wang et al., 2023). These models enable the system to generalize well across new and unseen DDoS variants, further reinforcing its long-term viability.

Crucially, this solution also addresses the resource constraints faced by many election commissions. It does not require physical deployment or expensive subscription models. As a lightweight system operating on existing cloud infrastructure, it democratizes access to enterprise-grade security. This accessibility is especially valuable in low-income or politically unstable regions where infrastructure investment is difficult, but where election integrity is vital. The ability to maintain system uptime and voter access during elections can enhance public trust and reduce the

political impact of cyber disruptions (Kim & Kim, 2023).

In conclusion, the proposed adaptive filtering extension offers a meaningful improvement to Project Shield's existing framework. By extending coverage to L4 and using real-time behavioral analysis, it transforms a traditionally application-only defense into a full-stack DDoS mitigation architecture. Its cloud-native design, learning capabilities, and modular deployment structure make it both scalable and cost-effective for government agencies safeguarding voter services in increasingly hostile cyber environments.

2.4 Similar Systems

To better understand the technological landscape and validate the relevance of the proposed system, this section compares several existing DDoS mitigation systems that incorporate Layer 4 and Layer 7 filtering across various sectors. These systems employ methods such as edge-based filtering, anomaly detection with autoencoders, and adaptive machine learning models. Each solution presents unique strengths in scalability, detection accuracy, and cost-efficiency, while also facing certain limitations such as deployment complexity or resource dependency. The table below outlines these systems, methods, advantages, and weaknesses to contextualize the proposed approach.

Author (Year)	System	Method	Strengths	Limitations	Link
Yaegashi et al. (2020)	Light-weight Edge DDoS Mitigation for resource s-limited environments	Queue-based identification and drop of malicious flows at edge routers using randomized shuffling and minimal hardware requirements	Designed for low-resource edge devices; early-stage packet filtering before reaching core systems	Primarily addresses simple flooding; not suitable for complex multi-vector attacks or application-layer threats	https://arxiv.org/abs/2010.15786
Li et al. (2023)	Adaptive Distributed Filtering	Granular rule deployment via edge nodes, dynamically adapting based on traffic prefixes and behavior scoring	Real-time adaptation; scalable across large networks; minimal impact on legitimate traffic	Requires more complex deployment (SDN or edge orchestration); likely high initial learning curve for infra setup	https://arxiv.org/abs/2305.13296
Bårli et al. (2021)	VAE-based Anomaly	Variational Autoencoder to model legitimate traffic, then detect anomalies by	Good at identifying stealthy or unknown DDoS attacks; does	Training requires clean data; may have false positives if reconstruction	https://arxiv.org/abs/2105.06899

	Detection	reconstruction error	not rely on labeled datasets	error thresholds are not well-tuned	
Anley et al. (2024)	CNN + Transfer Learning Framework	Uses convolutional neural networks and transfer learning to classify traffic patterns adaptively across datasets	High classification accuracy; adapts to new network environments or domains via transfer learning	Heavy ML model; GPU-required training; may overfit if datasets differ significantly across domains	https://www.sciencedirect.com/science/article/pii/S0167404824002670?via=ihub

3. Methodology

This methodology outlines the research design used to evaluate the proposed lightweight, cloud-native DDoS filtering system aimed at improving the resilience of government voter platforms. The approach includes identifying relevant user groups, selecting an appropriate sampling strategy, and defining the tools for data collection. A mixed-methods strategy will be applied to gather both quantitative and qualitative data, ensuring a balanced and comprehensive analysis

3.1 Target User

The proposed system is intended for government agencies responsible for conducting and managing elections, particularly those overseeing digital voter platforms, voter registration systems, and public election dashboards. These include national election commissions, state IT departments, and cybersecurity units tasked with protecting government digital assets during high-risk periods.

Indirect users include civic organizations involved in voter education, international election observers, and infrastructure

providers that host voter platforms. Additionally, central government regulators in charge of cybersecurity policy may play a role in supporting system adoption. The application is designed with scalability in mind to support both large-scale government agencies and smaller municipal election offices, especially those with limited access to enterprise security tools.

3.2 Sampling Method

The research will use **purposive sampling**, selecting participants based on their roles and responsibilities in managing election systems or public digital infrastructure. These participants will likely include IT managers, cybersecurity officers, government election coordinators, and systems administrators.

The target sample will come from both **urban and semi-urban areas**, representing a diversity of deployment contexts from national-level agencies to local councils. Demographically, participants will be aged 25–55, reflecting typical roles in technical and policy-making positions. Gender diversity will be considered in selection to maintain inclusive perspectives.

In later phases, cluster sampling may be applied to segment responses by region or agency type, allowing comparison across different operational environments and resource availability.

3.3 Data Collection Method

To gain a comprehensive understanding of how government agencies manage DDoS threats and evaluate the practicality of implementing adaptive L4–L7 filtering mechanisms, a mixed-method approach will be employed. This includes both quantitative and qualitative data collection techniques: a structured questionnaire and semi-structured interviews. This combination enables pattern analysis across a broader sample while also capturing deeper insights into infrastructure readiness, budget constraints, and stakeholder expectations.

i) Structured Questionnaire

A digital questionnaire will be distributed to IT personnel, cybersecurity officers, and digital infrastructure managers working in government election agencies. This tool will be used to collect **quantitative data** on DDoS defense strategies, platform performance, and interest in anomaly-aware filtering systems. The questionnaire will include **closed-ended questions**, Likert-scale statements, and multiple-choice items to ensure ease of data processing and analysis.

Structured Questionnaire Items:

1. **Infrastructure Readiness:**
 - Does your election platform use a cloud-based hosting provider?
 - ☐ Yes ☐ No ☐ Partially
 - What type of DDoS protection does your platform currently have?
 - ☐ Hardware-based ☐ Cloud-based ☐ None ☐ Not Sure
 - Does your current setup include Layer 4 protection?
 - ☐ Yes ☐ No ☐ Not Sure
2. **DDoS Exposure and Experience:**
 - How frequently has your voter system encountered traffic surges or suspected DDoS attacks in the past 12 months?
 - ☐ Never ☐ Rarely ☐ Occasionally ☐ Frequently ☐ Always
 - During election periods, how would you rate the stability of your platform?
 - [1 = Highly Unstable, 5 = Very Stable]
3. **Perception of Adaptive Filtering:**
 - Automated real-time filtering would be a valuable addition to our DDoS mitigation strategy.
 - [Strongly Disagree – Disagree – Neutral – Agree – Strongly Agree]
 - I would support integrating a lightweight anomaly-based filter that doesn't require hardware changes.
 - [Strongly Disagree – Disagree – Neutral – Agree – Strongly Agree]
4. **Operational Considerations:**
 - What is your team's current capacity for managing DDoS incidents?
 - ☐ Dedicated 24/7 SOC ☐ Limited on-call staff ☐ Outsourced ☐ None
 - What would be your primary concern about implementing a cloud-native adaptive filtering tool?
 - ☐ Cost ☐ Complexity ☐ Data privacy ☐ Compatibility ☐ None

Figure 2: Survey Questions

ii) Semi-Structured Interviews

Semi-structured interviews will be conducted with a **targeted selection of senior stakeholders**, such as Chief Information Officers (CIOs), heads of national election commissions, cybersecurity consultants, and cloud platform engineers involved in public infrastructure. These interviews aim to collect **qualitative insights** into operational challenges, technology limitations, and perceptions regarding adaptive DDoS mitigation.

Unlike structured surveys, semi-structured interviews offer flexibility. The core questions remain consistent, but interviewers can probe deeper based on the participant's experience, allowing a richer understanding of each organization's context.

Semi-Structured Interview Questions:

1. **Infrastructure and DDoS Impact**
 - o Can you describe a recent situation where your election system was affected by a DDoS attack or abnormal traffic spike?
 - o What mitigation strategies were used, and how effective were they?
 - o How does your team currently distinguish between legitimate traffic spikes and attack traffic?
2. **Technology Adoption and Readiness**
 - o Are there internal discussions about upgrading your current DDoS protection framework?
 - o What are the biggest technical or logistical challenges you foresee in adopting a cloud-native filtering solution?
 - o Have you explored integrating real-time anomaly detection tools into your infrastructure? Why or why not?
3. **Perceived Value and Risk**
 - o What benefits would you expect from a system that uses behavioral traffic analysis to automate mitigation?
 - o Are there any concerns regarding cloud deployment, especially related to data privacy or compliance?
 - o How open would your agency be to testing a lightweight adaptive filtering layer during an election simulation or load test?
4. **Feedback on Proposed Features**
 - o Based on your experience, what features should an adaptive filtering system prioritize?
 - o Would your agency be willing to contribute anonymized traffic data to help improve detection models if privacy were guaranteed?

Figure 3: Semi-Structured Interview Questions

4. Conclusion

As more governments use digital platforms to run elections and give people access to vote, it is becoming increasingly important to make sure that these systems are always available and safe against DDoS attacks. This study has shown that current public-sector mitigation measures have a big flaw: services like Google Project Shield don't have Layer 4 protection. Attackers can still target APIs, voter registration portals, and transport-layer connections even when Layer 7 defenses are in place. This can cause significant service outages. These events can make it harder to get to important services and make people less sure that digital elections are fair and reliable.

To solve this problem, the study suggests adding a real-time, anomaly-aware adaptive filtering mechanism to existing mitigation systems. This filtering layer learns how normal traffic behaves when things are quiet, and when it sees something unusual, it automatically applies the right rules. Its modular, cloud-native design means it can work without any extra hardware, which makes it very valuable for election organizations who don't have a lot of resources. The system's automation also cuts down on the need for people to get involved, which helps organizations stay responsive during busy times like election days.

One of the system's key advantages is its flexibility and scalability. It can be simply integrated with present infrastructures and changed to match regional peculiarities or policy requirements. The filtering model becomes better over time thanks to machine learning and feedback loops. This helps it keep up with new and changing attack tactics. Its lightweight architecture also provides cost-efficiency while providing strong protection, making it a feasible choice for both small and big government networks.

Adding more advanced technology to this system in the future could make it even better. Federated learning could let several electoral organizations work together to improve their detection models without putting users' privacy at

risk. Blockchain-based recording could make it easier to audit and hold people accountable by keeping clear, tamper-proof records of mitigating activities. Predictive analytics could assist in finding trends of attacks before they happen. Zero-trust security frameworks could also make external access points and internal operations even safer. If it

keeps getting better, this idea might become the basis for a smarter and more flexible defense plan for protecting democratic processes in the digital age.

(3536 words)

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6. Appendix

Semi-Structured Interview Questions:

1. Infrastructure and DDoS Impact

- Can you describe a recent situation where your election system was affected by a DDoS attack or abnormal traffic spike?
- What mitigation strategies were used, and how effective were they?
- How does your team currently distinguish between legitimate traffic spikes and attack traffic?

2. Technology Adoption and Readiness

- Are there internal discussions about upgrading your current DDoS protection framework?
- What are the biggest technical or logistical challenges you foresee in adopting a cloud-native filtering solution?
- Have you explored integrating real-time anomaly detection tools into your infrastructure? Why or why not?

3. Perceived Value and Risk

- What benefits would you expect from a system that uses behavioral traffic analysis to automate mitigation?
- Are there any concerns regarding cloud deployment, especially related to data privacy or compliance?
- How open would your agency be to testing a lightweight adaptive filtering layer during an election simulation or load test?

4. Feedback on Proposed Features

- Based on your experience, what features should an adaptive filtering system prioritize?
- Would your agency be willing to contribute anonymized traffic data to help improve detection models if privacy were guaranteed?

Structured Questionnaire Items:**1. Infrastructure Readiness:**

- Does your election platform use a cloud-based hosting provider?
 - ☐ Yes ☐ No ☐ Partially
- What type of DDoS protection does your platform currently have?
 - ☐ Hardware-based ☐ Cloud-based ☐ None ☐ Not Sure
- Does your current setup include Layer 4 protection?
 - ☐ Yes ☐ No ☐ Not Sure

2. DDoS Exposure and Experience:

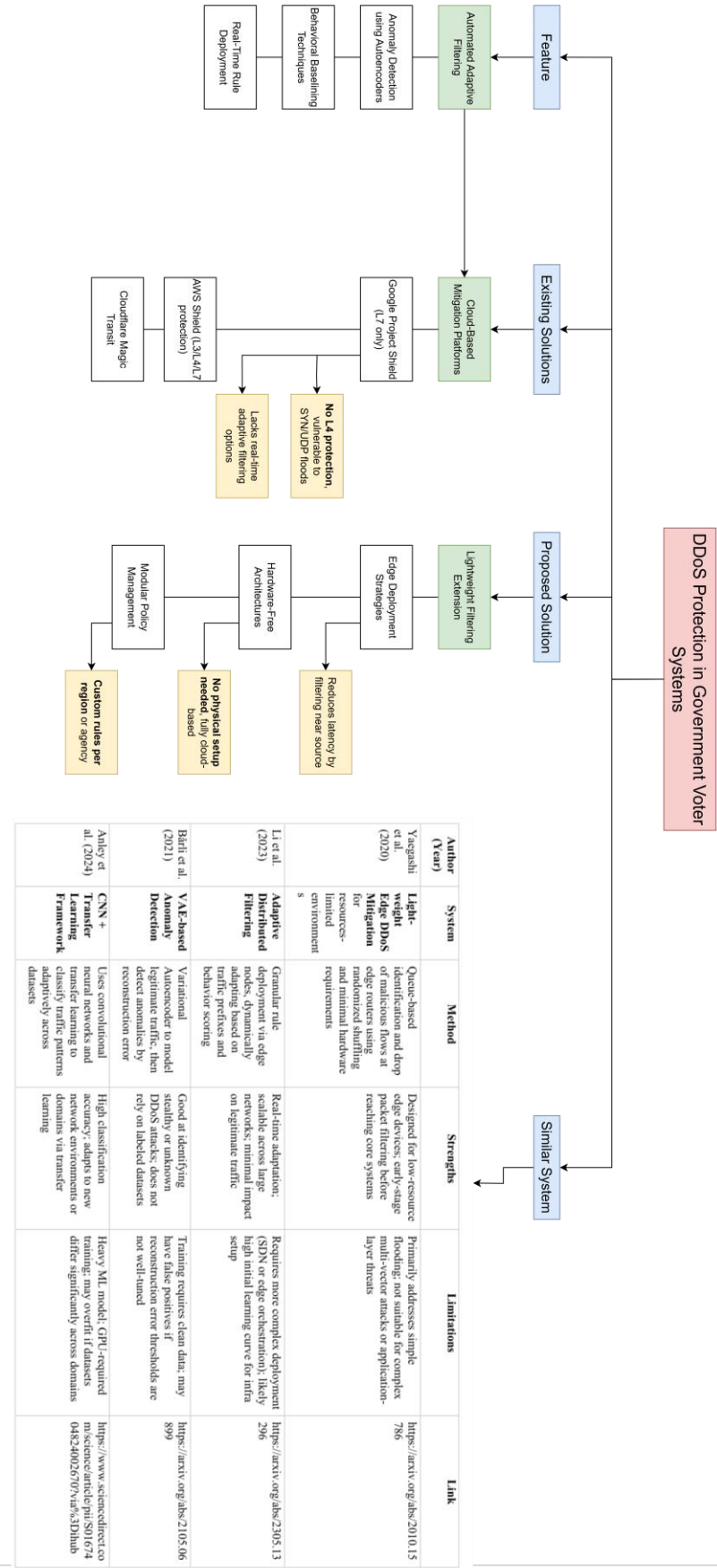
- How frequently has your voter system encountered traffic surges or suspected DDoS attacks in the past 12 months?
 - ☐ Never ☐ Rarely ☐ Occasionally ☐ Frequently ☐ Always
- During election periods, how would you rate the stability of your platform?
 - [1 = Highly Unstable, 5 = Very Stable]

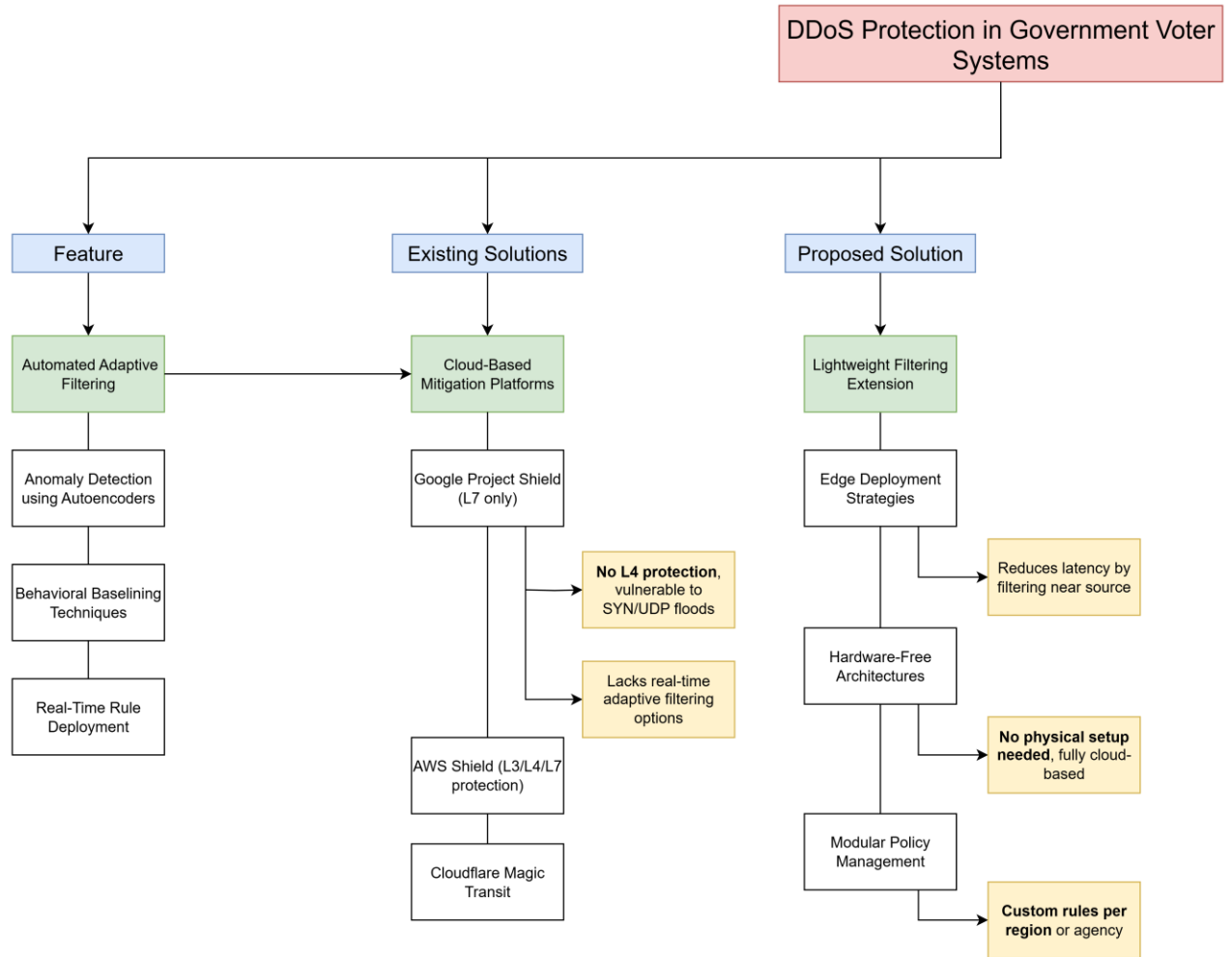
3. Perception of Adaptive Filtering:

- Automated real-time filtering would be a valuable addition to our DDoS mitigation strategy.
 - [Strongly Disagree – Disagree – Neutral – Agree – Strongly Agree]
- I would support integrating a lightweight anomaly-based filter that doesn't require hardware changes.
 - [Strongly Disagree – Disagree – Neutral – Agree – Strongly Agree]

4. Operational Considerations:

- What is your team's current capacity for managing DDoS incidents?
 - ☐ Dedicated 24/7 SOC ☐ Limited on-call staff ☐ Outsourced ☐ None
- What would be your primary concern about implementing a cloud-native adaptive filtering tool?
 - ☐ Cost ☐ Complexity ☐ Data privacy ☐ Compatibility ☐ None





Author (Year)	System	Method	Strengths	Limitations	Link
Yaegashi et al. (2020)	Light-weight Edge DDoS Mitigation for resource-limited environments	Queue-based identification and drop of malicious flows at edge routers using randomized shuffling and minimal hardware requirements	Designed for low-resource edge devices; early-stage packet filtering before reaching core systems	Primarily addresses simple flooding; not suitable for complex multi-vector attacks or application-layer threats	https://arxiv.org/abs/2010.15786
Li et al. (2023)	Adaptive Distributed Filtering	Granular rule deployment via edge nodes, dynamically adapting based on traffic prefixes and behavior scoring	Real-time adaptation; scalable across large networks; minimal impact on legitimate traffic	Requires more complex deployment (SDN or edge orchestration); likely high initial learning curve for infra setup	https://arxiv.org/abs/2305.13296
Bärli et al. (2021)	VAE-based Anomaly Detection	Variational Autoencoder to model legitimate traffic, then detect anomalies by reconstruction error	Good at identifying stealthy or unknown DDoS attacks; does not rely on labeled datasets	Training requires clean data; may have false positives if reconstruction error thresholds are not well-tuned	https://arxiv.org/abs/2105.06899
Anley et al. (2024)	CNN + Transfer Learning Framework	Uses convolutional neural networks and transfer learning to classify traffic patterns adaptively across datasets	High classification accuracy; adapts to new network environments or domains via transfer learning	Heavy ML model; GPU-required training; may overfit if datasets differ significantly across domains	https://www.sciencedirect.com/science/article/pii/S0167404824002670?via%3Dihub

References Drive Link:

https://drive.google.com/drive/folders/1KKOvdsCt5G2DjhrTxmvqR7XouW8Uzynl?usp=drive_link